# A CLOSED IMMERSION OF ALGEBRAIC SPACES IS PROPER

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The statement is one sentence; the proof is a two-step transfer through the representability of closed immersions and the scheme case.

*Editorial note.* The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

*Editorial note.* Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

*Allowed standard prerequisites (editorial).*

- A closed immersion of algebraic spaces is representable by definition. used at proof lines 8162-8163. Makes the transfer principle available.
- The general principle for properties of representable morphisms. Spaces, Lemma spaces-lemma-representable-transformations-property-implication, used at proof lines 8163-8165. Transfers properness from schemes to algebraic spaces.
- A closed immersion of schemes is proper. Morphisms, Lemma morphisms-lemma-closed-immersion-proper, used at proof lines 8165-8167. Is the scheme statement transferred.

*Source reference.* Original lemma 01W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2470–2470. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3883–3883. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *A closed immersion of algebraic spaces is a proper morphism of algebraic spaces.*

*Proof.* As a closed immersion is by definition representable this follows from Spaces, Lemma 02YO and the corresponding result for morphisms of schemes, see Morphisms, Lemma 01W5. □

*Date*: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.